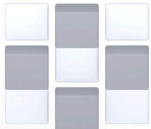
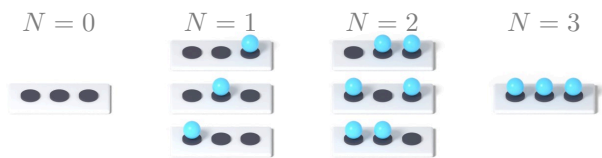
first quantization





$L = 3$ sites

second quantization, the full Fock space



$1 + 3 + 3 + 1 = 2^L = 8$ states